That is, $1 < \lim_{h\to 0} \frac{v^2}{c^2} < 1$. That is, $1 < \lim_{v\to c^-} \frac{v^2}{c^2} < 1$. By Sandwich theorem, $\lim_{v\to c^-} = 1$.

Therefore, $\lim_{v\to c^-}(m) \to \infty$.

That is, the mass becomes very very large (infinite) as $v \to c^-$.

The left hand limit is necessary. Otherwise as $v \to c^+$ makes $1 - \frac{v^2}{c^2} < 0$ and consequently we cannot find the mass.

Example 9.27

The velocity in ft/sec of a falling object is modeled by $r(t) = -\sqrt{\frac{32}{k}} \frac{1 - e^{-2t\sqrt{32k}}}{1 + e^{-2t\sqrt{32k}}}$, where k is a constant that depends upon the size and shape of the object and the density of the air. Find the limiting velocity of the object, that is, find $\lim_{t\to\infty} r(t)$.

Solution

$$\lim_{t\to\infty} r(t) = \lim_{t\to\infty} -\sqrt{\frac{32}{k}} \frac{1 - e^{-2t\sqrt{32k}}}{1 + e^{-2t\sqrt{32k}}}$$

$$= -\sqrt{\frac{32}{k}} \lim_{t\to\infty} \frac{1 - e^{-2t\sqrt{32k}}}{1 + e^{-2t\sqrt{32k}}}$$

$$= -\sqrt{\frac{32}{k}} \frac{(1-0)}{(1+0)} = -\sqrt{\frac{32}{k}} \text{ ft/sec.}$$

Example 9.28

Suppose that the diameter of an animal's pupils is given by $f(x) = \frac{160x^{-0.4} + 90}{4x^{-0.4} + 15}$, where x is the intensity of light and $f(x)$ is in *mm*. Find the diameter of the pupils with (a) minimum light (b) maximum light.

Solution

(a) For minimum light it is enough to find the limit of the function when $x \to 0^+$.

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{160x^{-0.4} + 90}{4x^{-0.4} + 15} = \lim_{x\to 0^+} \frac{160 + 90x^{0.4}}{4 + 15x^{0.4}}$$

$$= \frac{160}{4} = 40\text{mm}.$$

(b) For maximum light, it is enough to find the limit of the function when $x \to \infty$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{160x^{-0.4} + 90}{4x^{-0.4} + 15} = \frac{90}{15} = 6\text{mm}$$

That is, the pupils have a limiting size of 6mm, as the intensity of light is very large.

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